

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0614	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	G.Srivardhan	Reg. No.:	192425208
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type		Focus Area	Questions	Marks
Application-Based Questions		Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:			100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1.

A genome sequencing algorithm is modeled as:

$$\begin{aligned} &[\\ &T(n)=3T(n/2)+n \\ &] \end{aligned}$$

Apply the Master Theorem to determine the time complexity. Clearly identify the parameters and justify the case. Analyze how scalability affects large biological datasets.

Solution:

The given recurrence is:

$$\begin{aligned} &[\\ &T(n)=3T(n/2)+n \\ &] \end{aligned}$$

It matches the Master Theorem form:

$$[$$

$$T(n) = aT(n/b) + f(n)$$

$$]$$

Parameters:

- $(a=3)$
- $(b=2)$
- $(f(n)=n)$

Compute:

$$[$$

$$n^{\log_b a} = n^{\log_2 3} \approx n^{1.585}$$

$$]$$

Since

$$[$$

$$f(n) = n = O(n^{1.585 - \epsilon})$$

$$]$$

for some $(\epsilon > 0)$, Master Theorem Case 1 applies.

Therefore,

$$[$$

$$\boxed{T(n) = \Theta(n^{\log_2 3}) = \Theta(n^{1.585})}$$

$$]$$

Scalability Analysis

- The algorithm has sub-quadratic time complexity ($\Theta(n^{1.585})$), making it more efficient than $(O(n^2))$ algorithms.
- As genome size increases, execution time grows at a moderate rate, allowing efficient processing of large DNA sequences.
- The divide-and-conquer approach supports parallel execution, improving performance on multicore and distributed systems.
- Hence, the algorithm is suitable for analyzing large biological datasets such as whole-genome sequencing while maintaining good scalability.

Final Answer:

- Parameters: $(a=3,; b=2,; f(n)=n)$

- Master Theorem Case: Case 1
- Time Complexity: $\boxed{\Theta(n^{1.585})}$
- Scalability: Efficient for large genome datasets due to sub-quadratic growth and parallel processing capability.

Q 2

A distributed computing system schedules tasks using the recurrence:

$$\begin{bmatrix} T(n) = 7T(n/2) + n \end{bmatrix}$$

Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how this affects system performance as the number of tasks increases. Interpret efficiency using asymptotic notation.

Solution:

The given recurrence is:

$$\begin{bmatrix} T(n) = 7T(n/2) + n \end{bmatrix}$$

It matches the Master Theorem form:

$$\begin{bmatrix} T(n) = aT(n/b) + f(n) \end{bmatrix}$$

Step 1: Identify the Parameters

- $a = 7$
- $b = 2$
- $f(n) = n$

Step 2: Compute $(n^{\log_b a})$

$$\begin{bmatrix} n^{\log_2 7} \approx n^{2.807} \end{bmatrix}$$

Step 3: Compare $(f(n))$ with $(n^{\log_b a})$

Since

$$\begin{aligned} &[\\ f(n) &= n = O(n^{\{2.807 - \epsilon\}}) \\ &] \end{aligned}$$

for some ($\epsilon > 0$), Master Theorem Case 1 applies because ($f(n)$) grows asymptotically slower than ($n^{\log_2 7}$).

Step 4: Time Complexity

By Master Theorem Case 1,

$$\begin{aligned} &[\\ \boxed{T(n)} &= \Theta(n^{\log_2 7}) = \Theta(n^{\{2.807\}}) \\ &] \end{aligned}$$

System Performance Analysis

- The algorithm creates 7 recursive subproblems at each level, causing the number of computations to grow rapidly.
- As the number of tasks increases, execution time increases approximately as ($n^{\{2.807\}}$), which is significantly faster than linear growth but slower than cubic growth.
- Although distributed systems can execute tasks in parallel, the large number of recursive calls introduces scheduling and communication overhead.
- For very large task sets, performance decreases due to increased computation and synchronization costs.

Efficiency Using Asymptotic Notation

- Best Case: ($\Theta(n^{\{2.807\}})$)
- Average Case: ($\Theta(n^{\{2.807\}})$)
- Worst Case: ($\Theta(n^{\{2.807\}})$)

Final Answer

- Parameters: ($a=7$, $b=2$, $f(n)=n$)
- Master Theorem Case: Case 1

- Time Complexity: $\Theta(n^{2.807})$
- Scalability: As the number of tasks increases, execution time grows rapidly due to seven recursive calls at each level. The algorithm benefits from parallel execution but incurs higher scheduling overhead for large distributed systems. Thus, its efficiency is $\Theta(n^{2.807})$.

Code of Conduct

I G.Srivardhan(192425208) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						
Faculty In-charge		Course Coordinator		Head of Department		

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____